Water Level Monitoring System V1.0.1

Unified Communication Protocol Specification

Document Version: 1.0.1

Target Platform: ThingsBoard / Thingstack / IoT Gateway

1. Introduction

This document defines the data structures for the Water Level Monitoring System, covering Uplink (Device to Cloud) and Downlink (Cloud to Device) communications. It provides a unified guide for both **JSON-based (Cellular/WiFi)** and **Hex-encoded (LoRaWAN)** formats.

2. General Standards

2.1 Cellular/WiFi (JSON)

- **Format:** Standard UTF-8 JSON.
- **Identification:** Commands are identified by the "command" key.

2.2 LoRaWAN (Hex)

- **Endianness:** Little-Endian.
- **Packet Structure:**
  - **Start Byte:** F9 (1 byte).
  - **Direction:** 01 for Uplink, 02 for Downlink (1 byte).
  - **Size:** Size of the actual data payload without header and terminator in hex (1 byte).
  - **Terminator:** Newline character 0A (1 byte).

3. Uplink Packets (Device → ThingsBoard/Thingstack)

3.1 STARTUP_MESSAGE:

Sent immediately after a device boot cycle.

- **JSON Fields:**
  {"command":"STARTUP_MESSAGE","device_id":"{thingsboard_device_id}","reboot_reason":"software_reset","device_power_source":"mains","device_battery_percentage":97,"epoch_time":234567890,"sampling_frequency":30,"wifi_enabled":1,"mac_id":"12-34-56-78-90-12","connected_ap_ssid":"abc123","connected_ap_password":"abc123","ip_address":"192.168.0.1","cellular_enabled":1,"imei":"1567890","ccid":"1234567890","signal_strength":-10,"lorawan_enabled":1,"device_eui":"12-34-56-78-90","join_eui":"12-34-56-78-90","gps_enabled":1,"lbs_enabled":1,"hardware_version":"1.0.1","firmware_version":"1.0.1","reserved_1":"reserved_1_value","reserved_2":"reserved_2_value","reserved_3":"reserved_3_value"}
- **LoRaWAN Structure:**

  [F9:1B][Direction:1B][Size:1B][A1:1B][Thingstack_DevEUI:8B][RebootReason:1B][PowerSource:1B][Batt%:1B][Epoch:4B][SamplingFreq:2B][GPSEnabled:1B][LBSEnabled:1B][0A:1B]
- **Example Hex:** F90114A10901202600000001050164118968693C0000010A

3.2 HEALTH:

Periodic diagnostic heartbeat.

- **JSON Fields:**

  {"command":"HEALTH","device_id":"{thingsboard_device_id}","device_start_time":1234567890,"device_power_source":"mains","device_battery_percentage":97,"epoch_time":234567890,"sampling_frequency":30,"wifi_enabled":1,"mac_id":"12-34-56-78-90-12","connected_ap_ssid":"abc123","connected_ap_password":"abc123","ip_address":"192.168.0.1","cellular_enabled":1,"imei":"1567890","ccid":"1234567890","signal_strength":10,"lorawan_enabled":1,"device_eui":"12-34-56-78-90","join_eui":"12-34-56-78-90","gps_enabled":1,"lbs_enabled":1,"hardware_version":"1.0.1","firmware_version":"1.0.1","reserved_1":"reserved_1_value","reserved_2":"reserved_2_value","reserved_3":"reserved_3_value"}
- **LoRaWAN Structure:**

  [F9:1B][Direction:1B][Size:1B][A2:1B][Thingstack_DevEUI:8B][StartTime:4B][PowerSource:1B][Batt%:1B][Epoch:4B][SamplingFreq:2B][GPSEnabled:1B][LBSEnabled:1B][0A:1B]
- **Example Hex:** F90117A20901202600000001118968690164118968693C0000010A

3.3 DATA_PAYLOAD:

Primary telemetry packet for water level reporting.

- **JSON Fields:**

  {"command":"DATA_PAYLOAD","device_id":"{thingsboard_device_id}","device_power_source":"mains","device_battery_percentage":98,"sample_packet_number":1234,"epoch_time":234567890,"sampling_frequency":30,"water_level":9753}

- **LoRaWAN Structure:**

  [F9:1B][Direction:1B][Size:1B][A9:1B][Thingstack_DevEUI:8B][PowerSource:1B][Batt%:1B][PacketNo:4B][Epoch:4B][Freq:2B][WaterLevel:2B][0A:1B]

- **Example Hex:** F90117A9090120260000000101644433221144332211221119260 00A

3.4 SET_CONFIG_ACK:

Confirms the application of tank and radio settings.

- **JSON Fields:**
  {"command":"SET_CONFIG_ACK","device_id":"{thingsboard_device_id}","epoch_time":234567890,"sampling_frequency":30,"max_tank_capacity":300000,"tank_length":10,"tank_width":10,"tank_height":6,"wifi_enabled":0,"cellular_enabled":0,"lorawan_enabled":1,"gps_enabled":0,"lbs_enabled":0}

- **LoRaWAN Structure:**
  [F9:1B][Direction:1B][Size:1B][A3:1B][Thingstack_DevEUI:8B][Epoch:4B][Freq:2B][MaxCap:4B][Len:1B][Width:1B][Height:1B][WiFi:1B][Cell:1B][LoRa:1B][GPS:1B][LBS:1B][0A:1B]

- **Example Hex:** F9011BA3090120260000000111896869 3C00E09304000A0A0600000100000A

3.5 GET_CONFIG_ACK:

Returns current device settings to the platform.

- **JSON Fields:**
  {"command":"GET_CONFIG_ACK","device_id":"{thingsboard_device_id}","epoch_time":234567890,"sampling_frequency":30,"max_tank_capacity":300000,"tank_length":10,"tank_width":10,"tank_height":6,"wifi_enabled":0,"cellular_enabled":0,"lorawan_enabled":1,"gps_enabled":0,"lbs_enabled":0}

- **LoRaWAN Structure:**
  [F9:1B][Direction:1B][Size:1B][A4:1B][Thingstack_DevEUI:8B][Epoch:4B][Freq:2B][MaxCap:4B][Len:1B][Width:1B][Height:1B][WiFi:1B][Cell:1B][LoRa:1B][GPS:1B][LBS:1B][0A:1B]

- **Example Hex:** F9011BA4090120260000000111896869 3C00E09304000A0A0600000100000A

3.6 CONFIGURE_APN_ACK (Command: A5)

Confirms that the cellular APN has been updated.

- **JSON Fields:**
  {"command":"CONFIGURE_APN_ACK","device_id":"{thingsboard_device_id}","epoch_time":234567890,"apn":"jiociot2"}

- **LoRaWAN Structure:**
  [F9:1B][Direction:1B][Size:1B][A5:1B][Thingstack_DevEUI:8B][Epoch:4B][APN:16B][0A:1B]

- **Example Hex:** F9011DA50901202600000001118968690000000000000006A696F63696F74320A

3.7 REBOOT_ACK (Command: A6)

Confirms the receipt of the reboot command.

- **JSON Fields:**
  {"command":"REBOOT_ACK","device_id":"{thingsboard_device_id }"}

- **LoRaWAN Structure:**
  [F9:1B][Direction:1B][Size:1B][A6:1B][Thingstack_DevEUI:8B][0A:1B]

- **Example Hex:** F90109A609012026000000010A

3.8 RESET_TO_FACTORY_DEFAULTS_ACK (Command: A7)

Confirms all settings have been wiped to defaults.

- **JSON Fields:**

  {"command":"RESET_TO_FACTORY_DEFAULTS_ACK","device_id":"{thingsboard_device_id}"}

- **LoRaWAN Structure:**

  [F9:1B][Direction:1B][Size:1B][A7:1B][Thingstack_DevEUI:8B][0A:1B]

- **Example Hex:** F90109A709012026000000010A

3.9 TURN_OFF_EDGE_DEVICE_ACK (Command: A8)

Confirms the device is entering a shutdown or deep sleep state.

- **JSON Fields:**

  {"command":"TURN_OFF_EDGE_DEVICE_ACK","device_id":"{thingsboard_device_id}"}

- **LoRaWAN Structure:**

  [F9:1B][Direction:1B][Size:1B][A8:1B][Thingstack_DevEUI:8B][0A:1B]

- **Example Hex:** F90109A809012026000000010A

3.10 SET_LORAWAN_CONFIG_ACK (Command: AA)

Confirms the update of LoRaWAN network credentials.

- **JSON Fields:**

  {"command":"SET_LORAWAN_CONFIG_ACK","device_id":"{thingsboard_device_id}","epoch_time":1767370190,"lorawan_app_key":"237E7DE71BFC536FC30F18A02B925DE6","lorawan_nwk_key":"237E7DE71BFC536FC30F18A02B925DE6","lorawan_app_session_key":"237E7DE71BFC536FC30F18A02B925DE6","lorawan_nwk_session_key":"237E7DE71BFC536FC30F18A02B925DE6","lorawan_device_eui":"983ADC45238E89A3","lorawan_join_eui":"0101010101010101","lorawan_device_address":"013070E5"}

- **LoRaWAN Structure:**
  [F9:1B][Direction:1B][Size:1B][AA:1B][Thingstack_DevEUI:8B][Epoch:4B][NewAppKey:16B][NewNwkKey:16B][NewAppSKey:16B][NewNwkSKey:16B][NewDevEUI:8B][NewJoinEUI:8B][NewDevAddr:4B][0A:1B]
- **Example Hex:**
  F90161AA09012026000000001118968691D72EDF9392A91FC4A4F0C01AA133E191D72EDF9392A91FC4A4F0C01AA133E191D72EDF9392A91FC4A4F0C01AA133E191D72EDF9392A91FC4A4F0C01AA133E190901202600000001010101010101010101332211000A

3.11 CONFIGURE_MQTT_ENDPOINT_ACK

Confirms MQTT broker settings (supported for Cellular/WiFi only).

- **JSON Fields:**
  {"command":"CONFIGURE_MQTT_ENDPOINT_ACK","device_id":"{thingsboard_device_id}","epoch_time":234567890,"mqtt_endpoint":"devices.antzsystems.com","mqtt_endpoint_port_number":1883,"username":",djvkajdc","password":"SKDjcnksC","uplink_topic":"v1/devices/me/telemetry/swjefcbwksc","downlink_topic":"v1/devices/me/attributes/swjefcbwksc"}

3.12 CONFIGURE_MQTT_ENDPOINT_NACK

Rejects MQTT broker settings (supported for Cellular/WiFi only).

- **JSON Fields:**
  {"command":"CONFIGURE_MQTT_ENDPOINT_NACK","device_id":"{thingsboard_device_id}","epoch_time":234567890,"mqtt_endpoint":"devices.antzsystems.com","mqtt_endpoint_port_number":1883,"username":",djvkajdc","password":"SKDjcnksC","uplink_topic":"v1/devices/me/telemetry/swjefcbwksc","downlink_topic":"v1/devices/me/attributes/swjefcbwksc"}

4. Downlink Packets (ThingsBoard/Thingstack → Device)

4.1 SET_CONFIG:

Configures tank dimensions and radio status.

- **JSON Fields:**
  {"command":"SET_CONFIG","device_id":"{thingsboard_device_id}","epoch_time":234567890,"sampling_frequency":30,"max_tank_capacity":300000,"tank_length":10,"tank_width":10,"tank_height":6,"wifi_enabled":0,"cellular_enabled":0,"lorawan_enabled":1,"gps_enabled":0,"lbs_enabled":0}

- **LoRaWAN Structure:**
  [F9:1B][Direction:1B][Size:1B][B1:1B][Thingstack_DevEUI:8B][Epoch:4B][Freq:2B][MaxCap:4B][Len:1B][Width:1B][Height:1B][WiFi:1B][Cell:1B][LoRa:1B][GPS:1B][LBS:1B][0A:1B]

- **Example Hex:** F9021BB1090120260000000111896869 3C00E09304000A0A0600000100000A

4.2 GET_CONFIG:

Requests current device parameters.

- **JSON Fields:**

  {"command":"GET_CONFIG","device_id":"{thingsboard_device_id}"}

- **LoRaWAN Structure:**

  [F9:1B][Direction:1B][Size:1B][B2:1B][Thingstack_DevEUI:8B][0A:1B]

- **Example Hex:** F90209B209012026000000010A

4.3 CONFIGURE_APN:

Sets the Access Point Name for cellular connectivity.

- **JSON Fields:**

  {"command":"CONFIGURE_APN","device_id":"{thingsboard_device_id}","epoch_time":234567890,"apn": "jiociot2"}

- **LoRaWAN Structure:**
  [F9:1B][Direction:1B][Size:1B][B3:1B][Thingstack_DevEUI:8B][Epoch:4B][APN:16B][0A:1B]

- **Example Hex:** F9021DB309012026000000011189686 96A696F63696F74320000000000000000A

4.4 REBOOT:

Triggers a remote hardware restart.

- **JSON Fields:**

  {"command":"REBOOT","device_id":"{thingsboard_device_id}"}

- **LoRaWAN Structure:**

  [F9:1B][Direction:1B][Size:1B][B4:1B][Thingstack_DevEUI:8B][0A:1B]

- **Example Hex:** F90209B409012026000000010A

4.5 RESET_TO_FACTORY_DEFAULTS:

Instructs the device to wipe all user settings.

- **JSON Fields:**

  {"command":"RESET_TO_FACTORY_DEFAULTS","device_id":"{thingsboard_device_id}"}

- **LoRaWAN Structure:**

  [F9:1B][Direction:1B][Size:1B][B5:1B][Thingstack_DevEUI:8B][0A:1B]

- **Example Hex:** F90209B509012026000000010A

4.6 TURN_OFF_EDGE_DEVICE:

Commands the device to enter a powered-down or idle state.

- **JSON Fields:**

{"command":"TURN_OFF_EDGE_DEVICE","device_id":"{thingsboard_device_id}"}

- **LoRaWAN Structure:**

  [F9:1B][Direction:1B][Size:1B][B6:1B][Thingstack_DevEUI:8B][0A:1B]

- **Example Hex:** F90209B609012026000000010A

4.7 SET_LORAWAN_CONFIG:

Updates security keys and IDs for the LoRaWAN stack.

- **JSON Fields:**
  {"command":"SET_LORAWAN_CONFIG","device_id":"{thingsboard_device_id}","epoch_time":234567890,"lorawan_app_key":"237E7DE71BFC536FC30F18A02B925DE6","lorawan_nwk_key":"237E7DE71BFC536FC30F18A02B925DE6","lorawan_app_session_key":"237E7DE71BFC536FC30F18A02B925DE6","lorawan_nwk_session_key":"237E7DE71BFC536FC30F18A02B925DE6","lorawan_device_eui":"983ADC45238E89A3","lorawan_join_eui":"0101010101010101","lorawan_device_address":"013070E5"}

- **LoRaWAN Structure:**
  [F9:1B][Direction:1B][Size:1B][B7:1B][Thingstack_DevEUI:8B][Epoch:4B][NewAppKey:16B][NewNwkKey:16B][NewAppSKey:16B][NewNwkSKey:16B][New_Thingstack_DevEUI:8B][NewJoinEUI:8B][NewDevAddr:4B][0A:1B]

- **Example Hex:**
  F90261B709012026000000011189686934FBA5B1B48E84A8889CE9CDDBC8A56E34FBA5B1B48E84A8889CE9CDDBC8A56E34FBA5B1B48E84A8889CE9CDDBC8A56E34FBA5B1B48E84A8889CE9CDDBC8A56E20012026000000010101010101010101332211000A

4.8 CONFIGURE_MQTT_ENDPOINT

Sets broker details and topics (Cellular/WiFi only).

- **JSON Fields:**
  {"command":"CONFIGURE_MQTT_ENDPOINT","device_id":"{thingsboard_device_id}","epoch_time":234567890,"mqtt_endpoint":"devices.antzsystems.com","mqtt_endpoint_port_number":1883,"username":",djvkajdc","password":"SKDjcnksC","uplink_topic":"v1/devices/me/telemetry/swjefcbwksc","downlink_topic":"v1/devices/me/attributes/swjefcbwksc"}

5. Enumerations & Constants

Reboot Reasons:

0: Unknown

1: Low Power

2: Window Watchdog

3: Independent Watchdog

4: Software Reset

5: Power-On / Power-Down

6: External Pin

7: Brown-Out

Power Source:

0: Battery

1: Mains / DC

Water Level Scaling:

Transmitted as Integer. Value = Percentage x 100.

Example: 9753 (Dec) or 2619 (Hex, little-endian as 1926 in byte stream) = 97.53%